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M 410 PRESSURIZED AIR SYSTEM

1. GENERAL

This Section of the Specification describes the technical requirements for a Pressurized Air System (Service and Instrument Air System) and covers the minimum requirements for design engineering, manufacture, erection and inspection and testing. The Specifications of Part 4 entitled, Technical Requirements M-001, M-002, M-004, M-005 also includes Technical Requirements that are applicable to the Service and Instrument Air System covered by this Section of the Specification. In the event of any conflict between the Requirements of the listed specifications and the requirements of this Section then the latter shall take precedence.

The Bidder shall provide

- One (1) dedicated Pressurized Air Systems (Service Air and Instrument Air System) for Block 2 HRSG , Steam Turbine and their auxiliary systems
- One (1) dedicated Pressurized Air Systems (Service Air and Instrument Air System) for Block 3 & 4 HRSG , Steam Turbine and their auxiliary systems

The Service and Instrument Air System equipment shall be complete in every respect and suitable for safe reliable and continuous operation.

1.1 System Description

To cover the instrument air demand and the service air demand of Muara Tawar Combined Cycle Power Plant compressor skids, receiver and distribution systems complete with all necessary ancillaries shall be installed.

The main objectives of the compressed air systems are:

- Generation of compressed air in one centralized station
- Common production of service and instrument air
- Different service and instrument air networks
- Priority of dried instrument air

2. SCOPE OF WORK

This clause sets out the scope of the equipment and services as well as specially requested supplies and services but not excluding other necessary components and services not mentioned.

The Bidder shall provide and supply :

- one (1) complete Service and Instrument Air System for Block 2
- one (1) complete Service and Instrument Air System for Block 3
- **one (1) complete Service and Instrument Air System for Block 4**

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of Steam Turbine/HRSG and associated equipment, including all necessary components, spare parts and services.

The scope of supply and services shall include the design, manufacture, construction, factory testing, packing for transport, transport and delivery to site, unloading at site, storing, complete erection, testing at site, painting, commissioning, putting into normal operation, tie-in with existing facilities, reliability and performance testing of Service and Instrument Air System and associated equipment in accordance with Part 4, Subsection 1.2 “Summary of Work”.

Each system shall include but not be limited to the following :

- 2 x 100% motor driven compressors screw oil free type.
- Intercoolers, after coolers, moisture separators, drains and trays.
- Cooling System.
- Complete air distribution pipework with pipes, valves, anchor bolts and supports to the consumers.
- Remote and manual control and monitoring system.
- Inlet filters.
- Blow off Silencers
- 2 (two) Air receiver.
- 2 (two) x 100 % Instrument Air Dryer
- complete detailed labeling of all installations
- All other items which are necessary for the proper performance of the Service Air System & Instrument Air System.

- consumables such as first fill of lubricants and greases and any addition required up to the date of Take Over
- set of special tools and equipment for maintenance, inspection and repair
- one portable hydrometer
- all standard equipment and accessories which are normally included in the supply schedule but which are not separately listed
- detail design and calculation including influence on existing installation
- participation in all implementation discussions
- Training, inspection and supervision services by the Manufacturer of equipment
- Spare parts as specified
- complete documentation as specified

3. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The codes and standards extracted from Part 4, Subsection 2.5 “Codes and Standards” or other equivalent international standards are specifically applicable to the design, manufacture and testing of the equipment included in this Specification.

AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for testing and Material

The Bidder shall also comply with Indonesian Codes and Regulations regarding the operation of pressure vessels.

4. DESIGN AND OPERATING CONDITIONS

The Service and Instrument Air System shall allow to follow all operation

conditions of the Muara Tawar Combined Cycle Power Plant without any problem. Suitable preselection mode of each compressor shall be provided.

4.1 Air Compressors

The delivery air pressure of the compressor shall be controlled within the range of 6.5 to 9 bar (g).

The control system of Pressurized Air System shall be provided in order to maintain the constant air pressure for the Instrument Air System by automatically supplying pressurized air for the Instrument Air System, beside of continue supplying for the Service Air System each for HRSG and steam turbine.

The pressured air system shall be provided for the following:

- Instrument Air to HRSG and other unit to Steam Turbine.
- Distribution system serving all parts of the plant.
- Hose connection for turbine and heat recovery steam generator area.
- Purge air for primary piping for instrumentation as required.

4.2 Air Receiver

The net capacity of air receiver shall be sufficient for 15 minutes supply during normal operation. The terminal pressure of the air receiver shall be 6.5 bar (g) minimum.

The air receivers shall have a sufficient capacity for the Service/Instrument air in the power plant. The air receiver will be located downstream of the after coolers.

Air receivers will be mounted vertically and shall be mounted on a suitable base. Receivers shall include pressure gauge, drain connection and safety valve in accordance with ASME Section VIII, Division I requirements, inlet and discharge connections of suitable size 600 mm diameter hinged, manhole, drain trap and valves, etc. Inlets and outlets shall have flanged connections. The Bidder shall provide DEPNAKER certificate for pressure vessels.

Air receivers shall be of all welded construction and shall be designed and constructed for 1.10 times the maximum air Compressor surge pressure in accordance with the applicable trade standards of the Compressed Air Institute and ASME Boiler and Pressure Vessel Code, Section VIII, Division 1.

Receivers shall have ASME “U” stamp.

The air receiver shall be provided with an automatic drain trap.

5. DETAILED REQUIREMENTS

5.1 Air Compressor

Compressor shall be manufacturer’s package of standard design and construction with all integral piping and controls factory installed.

The compressors shall be of the water cooled oil free rotary screw type. The compressors shall be directly electric motor driven.

Compressors shall be designed and constructed for the most severe service conditions under which the compressors will be required to operate. All features of design and construction and all materials shall be acceptable to the Owner.

The Bidder shall supply the compressors, drives, skids and all auxiliaries. The Bidder has the responsibility for the overall design and performance of the compressor, drives, skids, auxiliaries and all other equipment included in the package.

All parts of each compressor unit shall be designed to withstand stresses resulting from a full-voltage start of the motor.

Bidder shall be responsible for ensuring that each compressor and motor operate in dynamic balance as a unit without undue vibration.

The whole assembly of the skid mounted units shall be rigid and balanced to be suitable for operation on a concrete slab.

The compressors shall be of robust construction and suitable for either intermittent or continuous operation. They shall be free from vibrations under all conditions of load and shall operate satisfactorily when left unattended for long periods.

The equipment shall be of such a design so as not to allow oil or other materials, which may cause a fire, explosion or coking hazard, to enter the compressed air system.

No plastic pipes shall be allowed.

Cooling water shall be supplied by the closed cooling water system.

The compressors shall be provided with safety devices and automatic control of the delivered air pressure.

Unless motor drives are designed to start the compressor under load automatic start-up unloading (unattended by operator) shall be provided to enable automatic starting or acceleration after a temporary electric power failure.

The start-up of a compressor shall be prevented if the cooling water flow is not satisfactory.

Should the “ON” compressor trip, the preselected compressor in stand-by shall start automatically.

When the pressure in the air receiver rises above normal operating pressure by a predetermined margin, the compressors shall be automatically unloaded. If the air receiver pressure falls below a pre-set figure, the compressors shall pick up load.

A device shall be included that will automatically bring the compressor to rest if it continues to run in an unloaded condition for more than a predetermined time.

The inspection and testing of the compressor unit shall be performed in the presence of the Owner and the Bidder at the works of the manufacturer. The manufacturer shall furnish certified copies of hydrostatic and running test records. These shall comprise performance data for all final tests including any difficulties arisen. The tests shall be carried out in accordance with the manufacturer’s standard test procedures.

Access to valves for maintenance, shall be possible without disturbing pipework and ancillary equipment.

The design of the inlet and outlet air filters shall be suitable for operation for prolonged periods in a dust laden atmosphere. Replaceable type air filters shall be provided capable of removing dust particles of 5 microns and sized for the worst atmospheric conditions possible on the Site. The differential pressure gauge and alarm shall be fitted across the air filter.

A suitable silencer shall be fitted to each compressor suction.

A non-return valve and an isolating valve shall be provided at the discharge of each compressor.

The compressors shall be installed in noise absorption compartments of proved design easy to dismantle for maintenance.

Sound pressure levels shall be rated and measured according to the ISO standards "Noise assessment with respect to annoyance".

Precision sound level meters as described in IEC 179 shall be used for measuring sound pressure levels.

When the air compressor unit equipment is running, the maximum acceptable level shall not exceed 85 dB (A) at 1 m distance.

The acoustic measurements shall be carried out in accordance with ISO 3744.

Air compressor units shall be constructed with materials specifically chosen for the operating and ambient conditions.

Compressors shall operate satisfactorily when delivering varying quantities and shall be constructed to ensure continuity of service and ease of operation, inspection, cleaning and repair, under all prevailing climatic conditions.

Pressure lubrication system complete with a main shaft driven gear type pump, strainer, relief valve and lubricating oil cooler. The lubricating oil shall be water cooled with a thermostatic control valve for cooling water flow.

Each air compressor unit shall be complete including but not limited to:

- air compressor machine
- driving motor
- intercooler, aftercooler
- air inlet filter and silencer as appropriate
- moisture separator
- all integral pipework, drains, traps and valves
- lubricating system including first fill of oils and greases

- common bed plate, together with all holding down bolts, packers, skims and anti- foundation steelwork
- controls, instruments
- sound absorbtion casing
- local panel
- lifting tackle
- all other equipment for safe and efficient operation.

5.2 Bearings

General Requirements :

- All bearings shall be located and arranged to facilitate alignment, maintenance and easy replacement.
- Suitable provisions shall be made to prevent “throwing” of grease or oil from the bearings.
- Axial thrust of the compressor motor(s) shall be counterbalanced to prevent overloading of individual thrust bearings.

5.2.1 Sleeve Bearings

Sleeve bearings shall be of manufacturer’s standard design with forced or ring oil lubrication.

- Wearing surfaces shall be of Babbitt or other approved antifriction alloy.

5.2.2 Thrust Bearings

- Thrust bearings shall be easily adjustable.
- Bearings shall be either Kingsbury or Michell self-contained type or tapered land type.

5.3 Shaft Seals

Shaft seals shall be manufacturer’s standard design suitable for the service conditions specified.

Seals shall be of the ring or labyrinth type with air purge.

All components subject to rubbing and wear on contact shall be of materials that have been tested and proven suitable for the specified service conditions.

Labyrinth seals shall be protected from excess discharge pressure, where necessary, by means of suitable pressure breakdown chambers or other approved pressure reducing devices.

Shaft seals shall be pressurized to prevent entry of dirt.

Shaft seals shall be designed to allow starting of the compressor independent of outside air source.

5.4 Casing

Compressors shall be of the horizontal or vertical split case design, constructed to permit ease of access for inspection and removal of rotating elements and replacement of wearing parts without disconnecting the suction and discharge piping and without disturbing compressor alignment.

Lifting lugs or eyes shall be provided as required.

Compressor suction and discharge connections shall be flanged.

5.5 Base and Soleplates

Compressor and motor shall be bolted to a common base that shall extend under and beyond all parts of the compressor and motor. All outside surfaces of the base shall have a smooth and finished appearance.

Compressor and motor shall be mounted and pre-aligned in the factory. Provisions shall be made to allow realignment in the field.

5.6 Couplings and Guards

Flexible type couplings shall be provided between the motor and the compressor. Couplings shall be suitable for the service intended.

Couplings shall be designed to withstand the stresses produced by starting motor at full voltage.

Coupling guards and all other guards necessary for protection against external moving parts shall be provided on the compressor. Guards shall be of substantial sheet steel construction suitably braced and reinforced and of the totally enclosed type.

5.7 Intercoolers and After Coolers

Air-cooled type intercoolers and after coolers shall be provided for each compressor.

Intercoolers and after coolers, shall conform to the ASME Boiler and Pressure Vessel Code, Section VIII, Division I and shall be Code inspected and stamped with the Code symbol.

After cooler shall be furnished with a moisture separator and an automatic condensate trap.

Intercooler shall be furnished with a pressure gauge and an automatic condensate trap for removing moisture.

Cooling air for Intercoolers and after coolers shall be ducted from outdoors.

Oil cooler, intercooler and after cooler shall have ample capacity (based on a cleanliness factor of 85%) to cool either the oil or the air to the desired temperature when supplied with the proper amount of cooling air.

The intercooler condensate drains shall have a brass wedge type gate valve, with self-cleaning V-notch orifice. The after cooler shall be provided with a trap on the moisture separator section.

5.8 Pipework and Valves

Piping and piping specialties shall be as per Part 4, Technical Specification M-001 Piping.

Bidder shall furnish inlet butterfly valve, discharge check valve, automatic shutoff valve and unloading valves per the requirements of Technical Specification M-004 Valves plus associated control air piping from control panel to the valves.

Bidder shall furnish suitable metal expansion joints for the air inlet, discharge and blow off terminal connections.

All necessary pipework and valves for the air system shall be provided under this Contract. The system shall enable isolation of any of the plant items to be made without interruption of the air supply to any other plant item. Drains shall be provided with traps at all points where condensation may occur and be detrimental to operation of the plant.

Air mains shall be sloped in the direction of the air flow so that any condensation shall travel towards traps by both gravity and air flow. Automatic traps with manual by-pass arrangements shall be installed at regular intervals and at low points in the system.

In the service air system a priority valve shall be installed which shuts off the service air in case of unacceptable air pressure drop and gives priority to the instrument air system.

All branches shall be taken from the top of the main using large radius bends. Caution is required to prevent any back flow into the system.

Each branch shall be isolated by its own stop valve and be provided with a drain trap.

On the service air system, hose connection points shall be provided at no more than 20 meters intervals around the plant. Each point shall be fitted with an isolating valve and quick release coupling, to the approval of the Owner. Sufficient number of flexible hose wheels shall be provided and stored inside a cabinet at various section of the plant.

The whole of the service air pipework systems shall be fabricated from carbon steel, hot dip galvanized in accordance with BS 729 with screwed connections. Valves shall be made of stainless steel or bronze. Screwed bonnet valves shall not be used.

Site welding of the hot dip galvanized pipes shall not be permitted at site.

The whole of the instrument air pipework systems shall be hot dip galvanized similar to service air pipework. Valves shall be made of stainless steel in accordance with ASTM A 182 Grade F 316L.

Instead of galvanized piping system stainless steel systems made of ASTM A 182 Grade F 316L or similar would also be acceptable.

The air distribution system shall be designed as a ring main system and be sized on a rather liberal basis.

5.9 Filter and Silencer

A dry type, two-stage intake filter-silencer shall be furnished complete with housing and replaceable media for outdoor location. The air filter shall be suitable to handle at least 150% of compressor capacity. Air filter medium shall not contain any oil or other chemical that may volatilize and cause a fire or explosion hazard in conjunction with compressed air.

A suitable blow off silencer to reduce discharge noise on the unloaded compressor cycle shall be furnished.

5.10 Air Receivers

A minimum of one receiver shall be provided for each air compressor.

Air receivers shall be of ASTM A 516 steel fabricated construction, designed and tested in accordance with ASME VIII, DIV.1.

Air receivers shall be provided with the following fittings including but not limited to all of which shall be subject to approval:

- a pressure and temperature gauge having a dial of not less than 150 mm diameter and isolating valve
- a direct spring loaded safety valve
- drain valve and trap
- floor support brackets
- a connection with isolating valve to serve as an initiating point for remote control air pressures and local indication
- a connection shall be provided for on/off pressure switches with adjustable set points
- A manhole of minimum 600 mm diameter shall be provided in each receiver.

5.11 Air Dryer

The air dryers shall be used to dry the compressed air from the air receiver for

use in various power plant control devices, pneumatic transmitters and instruments.

The air drier shall dry the air from the air receiver outlet conditions to a guaranteed dew point temperature and at the operating pressure.

Two desiccant air dryer units shall be provided, sized to match the output of the pressurized air compressor.

The Bidder shall state in his tender the type of desiccant used and shall provide one spare charge of the desiccant for each air dryer.

Connection for the portable hygrometer (included in the scope) on outlet mains shall be provided.

Pressure drop through entire, unit, dryer filter combination, shall not exceed 0.3 bar at the design flow.

Mounting bases and supporting framework shall be of rugged steel construction

rigidly braced to prevent distortion or misalignment of the equipment during shipment, handling and operation. All equipment shall be assembled on a common base at the factory.

The dryer shall be provided with integral inert gas piping, thermal mass piping, air piping and electrical wiring with all necessary instruments.

The air dryer shall have an integral control and instrument panel or cabinet to house the cycle controls and all instruments.

The air dryer shall be furnished with an automatic bypass.

The automatic bypass valve shall open on low downstream pressure and shall activate an indicating light on the dryer instrument panel. The bypass valve shall remain open until manually closed.

The automatic bypass valve shall be furnished with provisions for manual operation in case of valve malfunction.

The automatic bypass valve shall fail open on loss of air or electrical signal.

Single Pole Double Throw (SPDT) electrically isolated contacts shall be provided for the use to indicate the open and closed position of the bypass valve. The rating of the contacts shall be as indicated in Part 4, Technical Specification E-001.

The air dryer shall be furnished complete with all interconnecting piping, instrument panel with inlet and outlet pressure gauges, exit air moisture indicator, valves or cocks, relief valves, controls, automatic and manual bypass valves, pressure switches, thermometers and other appurtenances, including failure and warning light and high humidity alarm contact and light.

Bypass valves shall be capable of passing the full capacity of the dryer with the same equivalent pressure drop.

5.12 After Filter

Bidder shall furnish after filter capable of properly filtering the full design flow of the air dryer. Filter shall be designed for the specified service and shall be acceptable to the Owner.

The filter shall be installed complete and ready for service as part of the package unit.

The filter shall be furnished with an upstream and downstream isolated valve.

5.13 Painting and Protective Coating

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and shall be painted as specified below.

All parts of the equipment, except the electrical apparatus, instruments etc.,

shall be finish painted with of manufacturer's standard paint suitable for the service.

Electrical apparatus, instruments etc., shall be painted in accordance with Bidder's standard practice for the equipment.

Exterior surfaces of the equipment that are subject to temperatures above 120°C shall be painted with heat-resistant paint, as approved.

Refer also to specifications S-420 and S-430.

No paint having an asphaltum base shall be used

Before shipment all exposed machined ferrous metal surfaces shall be coated with a suitable antirust compound. Inlet, outlet and all other orifices shall be blanked off or plugged.

6. PERFORMANCE TESTING & GUARANTEES

Refer to specification M-014 Inspection and Testing.

6.1 Shop Test

The Bidder shall perform a full load test on the first compressor in accordance with ASME PTC 10, compressor and exhausters " class I to determine the compressor rating for flow, pressure and surge point and to insure that the assembled package is properly balanced. Controls shall be factory adjusted. Certified copies of all shop tests performed shall be sent to the Owner.

All final shop testing to demonstrate compliance with the specification must be done with the project unit's own intercoolers, piping and lube oil system.

6.2 Field Test

After equipment and auxiliaries are operational the Bidder shall carry out a field test checking the compressor delivery pressure and the functioning of all equipment without abnormal vibration and to ascertain that the equipment is performing properly in every respect.

7. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

7.1 Pressurized Air System

Quality assurance/Quality control shall conform to the requirements of part 4, Subsection 2.7.

The quality control procedures so specified in QA-2-MTCCPP shall include but not be limited to the following:

- Welding
- Nondestructive testing
- Packaging and shipping
- Shop test procedures
- Cleaning and painting

The quality control documentation as specified shall include, but not be limited to the following

- Test reports
- Material certifications
- NDE Reports
- All other documentation required by applicable codes and standards.

8. INSTRUMENTATION AND CONTROLS

Compressor shall be furnished with controls and accessories as required for controlling each unit from no load to full load. Controls shall be designed so that either of two modes of control can be selected by means of a selector switch on Control panel. One mode shall be constant pressure and surge control, which controls the discharge pressure by modulating the inlet throttle valve and unloads at surge pressure. The other mode shall be intermittent load control, which loads and unloads the compressor between pressure limits. Unloaded operation below the surge point shall provide only sufficient flow to cool the machine at minimum horsepower.

8.1 Control Panel

Control Panel shall be integral with the compressor, in accordance with Part 4, Technical Specifications I-002 and E-001, and shall include the following:

Discharge air pressure and temperature gauges.

Warning and shutdown, alarm devices shall have two settings for the conditions listed below. One setting shall be for a warning alarm and the other shall be for tripping the compressor.

- High oil temperature.
- High air temperature (inter stage and discharge).
- Low oil pressure (one for stopping compressor and one for oil pump).
- Low water pressure.

Oil pressure gauge.

Any additional control devices required to provide the type of control described above.

Terminal contact for wiring of remote switches to allow starting and stopping of compressor from Main Control Room and terminals for remote common alarm light and common trip light.

High inlet filter differential pressure (alarm only).

8.2 Safety Interlock

Any required safety interlocks not specifically mentioned in this specification shall be included if Bidder feels they are necessary for safe and reliable operation.

8.3 Connections

Instrumentation connections necessary to monitor and verify the performance of the compressor in each stage shall be furnished.

8.4 Main Motor Tripping

All control devices which will trip the main motor or are used for interlocking with the main motor shall be furnished with two (2) double pole, double throw switches. One contact shall be used for the trip function. The other contact shall be used for annunciator.

8.5 System Protection

Each system circuit shall be properly protected with Circuit breakers or pull

out fuses.

8.6 Vibration Avoidance

Control devices shall be mounted in such a manner that vibration will not be transferred to these devices during compressor operation. Wiring to control devices shall be enclosed in- flexible, liquid tight conduits. Conduit shall be properly grounded.

8.7 Automatic Loading

Pressure switch to permit automatic loading of compressor on low air pressure when the compressor is running and unloaded via remote control switches shall be included.

8.8 Other Alarm Points

The Bidder shall provide alarm points for all conditions as required for safe operation of the equipment.

8.9 Air Flow Monitoring Equipment

The Bidder shall furnish air flow indicators for each compressor consisting of a flow meter and nozzle for installation in the discharge line with interconnection to the control panel.

8.10 Other Instrumentation

The Bidder shall furnish:

- Blow off valve position indicator lights.
- Pressure gauges at each stage outlet and each oil pump outlet and heat exchanger cooling water discharge header.
- After cooler discharge air high temperature alarm switches.
- An automatic cold start system which energizes the seal air system and pre-lube oil pump.
- Temperature gauges for each cooler outlet water pipe.
- A filter-silencer differential pressure switch and alarms for each filter stage.
- A visual pressure drop indicator with each intake filter-silencer
- Inlet butterfly valve position indicator mounted on the valve body, stem, or as approved by the Owner.
- Bearing thermocouple.

Start and stop of service air compressor shall be made possible remote operation from Central Control Desk in Main Control room.

Refer to Part 4, Technical Specification I-001 for the design requirements for the instrumentation and control devices.

9. ELECTRICAL SPECIFICATIONS

The compressor motor shall be in accordance with Part 4, Technical Specification E-211. Miscellaneous small motors shall be in accordance with Part 4, Technical Specification E-210.

The design of all miscellaneous electrical components shall be in accordance with Part 4, Specifications E- 001, I-001 and I-002.

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